Finite-dim

$$Z(g) = \int d^N\phi \; e^{-\frac{1}{2}\phi^t A \phi - g \sum_i \phi_i^4} \qquad = -S(\phi) \qquad \nearrow \langle \phi_i \phi_j \rangle_0 = A^{-1}_{ij}$$

$$\langle \phi_{i_1} \phi_{i_2} \cdots \phi_{i_n} \rangle = \frac{1}{Z(g)} \int d^N\phi \; e^{-\frac{1}{2}\phi^t A \phi - g \sum_i \phi_i^4} \phi_{i_1} \phi_{i_2} \cdots \phi_{i_n}$$

Going to a field in spacetime: labels i (discrete) $\rightarrow$ x (continuous)

$$\sum_i \rightarrow \int d^4x$$

$$-\frac{1}{2}\underbrace{\phi^t}_{\text{row}} \underbrace{A\phi}_{\text{column}} = -\frac{1}{2}\sum_i \phi_i (A\phi)_i$$

$$\int d^N\phi \rightarrow \int D\phi$$

operator

$$Z(g) = \int D\phi \; e^{-\int d^4x \left[\frac{1}{2}\phi(x)(K\phi)(x) + g\,\phi(x)^4\right]} \qquad (*)$$

Wick-rotated version

Bosonic field $\quad S[\phi] = \int d^4x \; \frac{1}{2}\left((\partial_t\phi)^2 - (\nabla\phi)^2 - m^2\right) + g\,\phi(x)^4$

Non-Wick Rotated

$$Z(g) = \int D\phi \; e^{i\int d^4x \; \frac{1}{2}\left((\partial_t\phi)^2 - (\nabla\phi)^2 - m^2\right) + g\,\phi(x)^4}$$

$t \rightarrow +it \qquad i\int d^4x \rightarrow -\int d^4x \quad , \quad (\partial_t\phi)^2 \rightarrow -(\partial_t\phi)^2$

$$Z(g) = \int D\phi \; e^{-\int d^4x \; \frac{1}{2}\left(-(\partial_t\phi)^2 - (\nabla\phi)^2 - m^2\right) + g\,\phi(x)^4}$$

<u>Note:</u> $\quad \int dt \; (\partial_t\phi)^2 = -\int dt \; \phi\, \partial_t^2 \phi \qquad \swarrow$ integration by parts

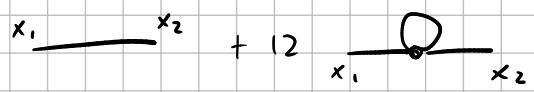
$$\int d^3\vec{x} \; (\nabla\phi)^2 = -\int d^3\vec{x} \; \phi\, \nabla^2\phi$$

$$Z(g) = \int D\phi \; e^{-\int d^4x \; \underbrace{\frac{1}{2}\left(\phi(\partial_t^2+\nabla^2)\phi - m^2\right) - g\,\phi(x)^4}}$$

looks like (*) with $K\phi = (\partial_t^2 + \nabla^2)\phi - m^2\phi$

$$\langle \phi(x_1) \cdots \phi(x_n) \rangle = \frac{1}{Z(g)} \int D\phi \; e^{-\int d^4x \; \cdots} \; \phi(x_1) \cdots \phi(x_n)$$

$$\langle \phi(x_1)\,\phi(x_2) \rangle = \langle \phi(x_1)\,\phi(x_2) \rangle_0 - 12\, g \int d^4x \, \langle \phi(x_1)\,\phi(x)\rangle_0 \, \langle \phi(x)\,\phi(x_1)\rangle_0 \, \langle \phi(x)\phi(x)\rangle_0$$

$=$ 

only difference in rules for diagrams: each vertex has an integral over a continuous label

what does $\langle \phi(x_1)\,\phi(x_2)\rangle_0$ look like?

in a sense $\langle \phi(x_1)\,\phi(x_2)\rangle_0 = K^{-1}(x_1, x_2)$?

"$\int d^4x \; K(x_1, x)\, K^{-1}(x, x_2) = \delta(x_1 - x_2)$"

for a matrix $(AA^{-1})_{ij} = \sum_k A_{ik} A^{-1}_{kj} = \delta_{ij}$

Green's function: $K\, G(x_1, x_2) = \delta(x_1 - x_2)$

$K = \partial_t^2 + \nabla^2 - m^2$ since K is translation invariant,

$G(x_1, x_2) = \tilde{G}(x_1 - x_2)$

$\rightarrow K\, G(x) = \delta(x) \rightarrow (\partial_t^2 + \nabla^2 - m^2)\, G(x) = \delta(x)$

switch to Fourier transform $G(x) = \int \frac{d^4p}{(2\pi)^2} e^{ip\cdot x} \hat{G}(p)$

$$\partial_{x_\mu} G(x) = \int \frac{d^4p}{(2\pi)^2} \partial_{x_\mu} e^{ip\cdot x} \hat{G}(p) = \int \frac{d^4p}{(2\pi)^2} \, i p_\mu \hat{G}(p) \, e^{ip\cdot x}$$

$$\delta(x) = \frac{1}{(2\pi)^2} \int \frac{d^4p}{(2\pi)^2} e^{ip\cdot x}$$

$$\partial_{x^\mu}\partial_{x^\nu} G(x) = -\int \frac{d^4p}{(2\pi)^2}\, p_\mu p_\nu\, \hat{G}(p)\, e^{ip\cdot x}$$

$$(\partial_t^2 + \nabla^2) G(x) = -\int \frac{d^4p}{(2\pi)^2}\, p^2\, \hat{G}(p)\, e^{ip\cdot x}$$

$$\rightarrow \int \frac{d^4p}{(2\pi)^2}\, \hat{G}(p)\left(-p^2 - m^2\right) e^{ip\cdot x} = \frac{1}{(2\pi)^2}\int \frac{d^4p}{(2\pi)^2}\, e^{ip\cdot x}$$

$$\Rightarrow \boxed{\hat{G}(p) = -\frac{1}{(2\pi)^2}\,\frac{1}{p^2+m^2}}$$

$$\boxed{G(x_1 - x_2) = -\lim_{\varepsilon\to 0^+}\int \frac{d^4p}{(2\pi)^4}\,\frac{e^{ip(x_1-x_2)}}{p^2+m^2-i\varepsilon}}$$

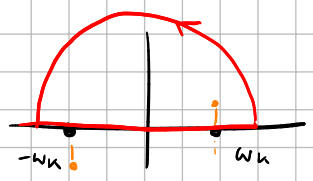
We are with rotating back $t \to -it$, $p_0 \to -i p_0$

$$p^2 \to -p_0^2 + \vec{p}^{\,2} = -p_\mu p^\mu \qquad\qquad p_0 t = -p_0 t$$

$$G(x_1 - x_2) = -i \lim_{\varepsilon\to 0^+}\int \frac{d^4p}{(2\pi)^4}\,\frac{e^{ip(x_1-x_2)}}{-p_0^2+\vec{p}^{\,2}+m^2-i\varepsilon} \qquad p\cdot x = p_0 x_0 - \vec{p}\cdot\vec{x}$$

$$= i \lim_{\varepsilon\to 0^+}\int \frac{d^4p}{(2\pi)^4}\,\frac{e^{ip\cdot(x_1-x_2)}}{+p_0^2-(\vec{p}^{\,2}+m^2)+i\varepsilon}$$

without ε, integral w.r.t. p_0 has 2 poles at $p_0 = \pm\underbrace{\sqrt{\vec{p}^{\,2}+m^2}}_{\omega_k}$



other approaches: $(p_0 + i\varepsilon)^2 - (p^2+m^2)$

<u>decide</u>: $\boxed{\langle \phi(x_1)\,\phi(x_2)\rangle_0 = G(x_1 - x_2)}$ analogous to $\langle \phi_i\,\phi_j\rangle_0 = A^{-1}_{ij}$

We often prefer to look at Feynman diagrams in momentum space

look at interaction term $\int d^4x\, g\, \phi(x)^4$

Fourier transform:

$$\varphi(p) = \int \frac{d^4x}{(2\pi)^2} e^{-ip\cdot x} \phi(x)$$

$$\phi(x) = \int \frac{d^4p}{(2\pi)^2} e^{+ip\cdot x} \varphi(p)$$

$$\int \frac{d^4x}{\cancel{(2\pi)^2}} e^{ip\cdot x} = \frac{1}{\cancel{(2\pi)^2}} \delta(p)$$

$$\int d^4x \, g \, \phi(x)^4 = \int d^4x \, g \int \frac{d^4p_1}{(2\pi)^2} \int \frac{d^4p_2}{(2\pi)^2} \int \frac{d^4p_3}{(2\pi)^2} \int \frac{d^4p_4}{(2\pi)^2} e^{i(p_1+p_2+p_3+p_4)\cdot x} \varphi(p_1)\varphi(p_2)\cdots\varphi(p_4)$$

$$g \int \frac{d^4p_1}{(2\pi)^2} \int \frac{d^4p_2}{(2\pi)^2} \int \frac{d^4p_3}{(2\pi)^2} \int \frac{d^4p_4}{(2\pi)^2} \underbrace{\delta(p_1+p_2+p_3+p_4)}_{\text{conservation of momentum}} \varphi(p_1)\varphi(p_2)\cdots\varphi(p_4)$$

What we want to do: instead of $\langle \phi(x_1) \cdots \phi(x_n) \rangle$

we want to look at $\langle \varphi(p_1)\varphi(p_2) \cdots \varphi(p_n) \rangle$

$$\langle \varphi(p)\varphi(q) \rangle = \int \frac{d^4x}{(2\pi)^2} \frac{d^4y}{(2\pi)^2} \langle \phi(x)\phi(y) \rangle e^{ip\cdot x} e^{iqy}$$

<u>Gaussian case:</u> $\langle \varphi(p)\varphi(q) \rangle_0 = \int \frac{d^4x}{(2\pi)^2} \frac{d^4y}{(2\pi)^2} \langle \phi(x)\phi(y) \rangle_0 e^{ip\cdot x} e^{iqy}$

$$= \int \frac{d^4x}{(2\pi)^2} \frac{d^4y}{(2\pi)^2} G(x-y) e^{ip\cdot x} e^{iqy}$$

$u = x - y$

$d^4u = d^4x$

$e^{ip\cdot x} = e^{ip(u+y)}$

$$= \int \frac{d^4u}{(2\pi)^2} \frac{d^4y}{(2\pi)^2} G(u) e^{ip\cdot u} e^{i(p+q)y}$$

$$= \frac{1}{(2\pi)^2} \delta(p+q) \int \frac{d^4u}{(2\pi)^2} G(u) e^{ip\cdot u}$$

Fourier transform of $G \rightarrow \hat{G}(p) = \frac{i}{(2\pi)^2} \frac{1}{p_0^2 - (\vec{p}^2 + m^2) + i\varepsilon}$

$$\langle \varphi(p) | \varphi(q) \rangle_0 = \frac{1}{(2\pi)^2} \underbrace{\delta(p+q)}_{\text{conservation of momentum}} \frac{i}{(2\pi)^2} \frac{1}{p_0^2 - (\vec{p}^2 + m^2) + i\varepsilon}$$

$$\langle \phi(x_1)\varphi(x_2)\rangle = \frac{Z(0)}{Z(g)} \sum_{k=0}^{\infty} \frac{(ig)}{k!} \langle \phi(x_1)\phi(x_2)\left(\int d^4x\, \phi(x)\right)^k \rangle_0$$

Wick rotation

in momentum space:

$$\langle \phi(x_1)\varphi(x_2)\rangle = \int \frac{d^4p}{(2\pi)^2}\frac{d^4q}{(2\pi)^2} \langle \varphi(p)\varphi(q)\rangle e^{ip\cdot x_1} e^{iq\cdot x_2}$$

replace $\phi(x_1)$ with $\int \frac{d^4p}{(2\pi)^2} \varphi(p) e^{ip\cdot x_1}$

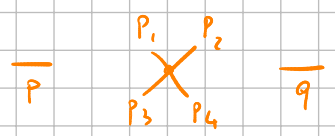
$$\langle \phi(x_1)\phi(x_2)\left(\int d^4x\, \phi(x)\right)^k \rangle_0$$

$$= \int \frac{d^4p}{(2\pi)^2}\frac{d^4q}{(2\pi)^2} e^{ip\cdot x_1} e^{iq\cdot x_2} \langle \varphi(p)\varphi(q)\left(\int d^4x\, \phi(x)\right)^k \rangle_0$$

$$\langle \varphi(p)\varphi(q)\rangle = \frac{Z(0)}{Z(g)} \sum_{k=0}^{\infty} \frac{(ig)}{k!} \langle \varphi(p)\varphi(q)\left(\int d^4x\, \phi(x)\right)^k \rangle_0$$

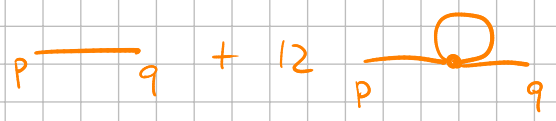
$$\int d^4x\, \phi(x)^4 = \int \prod_{i=1}^{4} \frac{d^4p_i}{(2\pi)^2}\, \delta\left(\sum_i p_i\right) \varphi(p_1)\varphi(p_2)\varphi(p_3)\varphi(p_4)$$

$$\langle \varphi(p)\varphi(q)\rangle = \frac{Z(0)}{Z(g)} \sum_{k=0}^{\infty} \frac{(ig)}{k!} \langle \varphi(p)\varphi(q)\left[\int \prod_{i=1}^{4} \frac{d^4p_i}{(2\pi)^2}\, \delta\left(\sum_i p_i\right) \varphi(p_1)\varphi(p_2)\varphi(p_3)\varphi(p_4)\right]^k \rangle_0$$



to first order

$$\langle \varphi(p)\varphi(q)\rangle = \langle \varphi(p)\varphi(q)\rangle_0 + 12ig \int \prod_{i=1}^{4} \frac{d^4p_i}{(2\pi)^2}\, \delta\left(\sum_i p_i\right) \langle \varphi(p)\varphi(p_1)\rangle_0 \langle \varphi(q)\varphi(p_4)\rangle_0 \langle \varphi(p_2)\varphi(p_3)\rangle_0$$



Rules:

- each vertex carries a factor of ig, and a "sum" over 4 internal momenta $p_1, p_2, \dots, p_4$ with $\sum_i p_i = 0$ → so a $\delta\left(\sum_i p_i\right)$

- each connection between two legs with momenta p, q is a propagator $\hat{G}(p)\,\delta(p+q)$